# Swachh Bharat Mission Evaluation

*Sanitation Access and Use in Telangana and Bihar*

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## Abstract

This paper evaluates the Swachh Bharat Mission by comparing Telangana and Bihar, taken here as a top-performing and a low-performing state on open defecation outcomes. It reviews state-level studies and uses NSS data to examine access to latrines, reasons for not using them, and the availability of water in latrines across social groups, gender and sector. The evidence suggests that access to a latrine does not guarantee sustained use. Bihar reports high receipt of Mission benefits but weaker outcomes, and trust, monitoring, social norms and water availability all shape whether latrines are used. The paper concludes that infrastructure, governance and social change have to work together.

## Introduction

India has made considerable progress in reducing open defecation through the Swachh Bharat Mission, launched in 2014 in both rural and urban areas. The government describes it as the world's largest sanitation campaign, and reports that only about 40% of households had access to toilets before it began (Department of Drinking Water and Sanitation, 2024). The Mission focuses on behavioural change, social inclusion and the safety of women, and it set out to make India Open Defecation Free (ODF). By 2 October 2019, the 150th birth anniversary of Mahatma Gandhi, more than 100 million individual household toilets had been built and more than 6 lakh villages had declared themselves ODF, in line with SDG target 6.2 (Ministry of Jal Shakti, 2023).

Open defecation nevertheless persists, and many villages still do not meet the basic conditions for sanitation. Phase II of the rural Mission (SBM-G) set the target of making all villages ODF Plus by 2025. According to figures released in September 2023, 4,43,964 villages had reached ODF Plus status: 2,92,497 were ODF Plus Aspiring villages, with arrangements for either solid or liquid waste management; 55,549 were ODF Plus Rising villages, with arrangements for both; and 96,018 were ODF Plus Model villages. The three categories as reported add up to 4,44,064, slightly above the stated total. Overall, 2,31,080 villages had arrangements for solid waste management and 3,76,353 for liquid waste management (Ministry of Jal Shakti, 2023).

Under the guidelines of SBM-Gramin, sanitation is a state subject, and states decide their own policy interventions. This paper therefore compares two states, Telangana and Bihar, which are taken as a top-performing and a low-performing state respectively on open defecation targets. It examines the determinants of sanitation in each state, explores why open defecation persists, and uses the two cases to understand the challenges and policy interventions needed to meet the Mission's targets.

## Reasons for Open Defecation

India's open defecation rates are far higher than those of other developing regions where people are poorer, literacy is lower and water is relatively scarcer. Drawing on survey and interview data from rural north India, Coffey et al. (2017) argue that government programmes have paid little attention to why so many rural Indians defecate in the open rather than use affordable pit latrines, and that the answer lies in the history of caste and untouchability.

The comparison is striking. Coffey et al. (2017) report that the 2011 Census found about 70% of rural Indian households without a toilet or latrine. By contrast, only about 35% of people in rural sub-Saharan Africa, 5% in rural Bangladesh and 2% in rural China defecate in the open (UNICEF and WHO, 2012, as cited in Coffey et al., 2017). The two measures differ, since lacking a latrine at home is not the same as defecating in the open, so the comparison is indicative rather than exact.

Open defecation needs to be addressed because it is not only a sanitation issue but is linked to poor health, especially in places with high population density. It spreads bacterial, viral and parasitic infections, including diarrhoea, polio, cholera and hookworm, and is an important cause of child stunting and infant death (Coffey et al., 2017).

Coffey et al. (2017) argue that the explanation lies less in material or educational deprivation than in beliefs, values and norms about purity, pollution, caste and untouchability, which lead many people to reject affordable pit latrines. These latrines require manual emptying. Upper-caste households tend to regard emptying a pit as impure and not their work, while lower-caste households refuse the task because of a long history of oppression. In this view, defecating in the open is seen as pure, whereas a latrine pit near the home is seen as a source of pollution. The same study also reports that people who receive government latrines often do not use them for defecation, and that households which build their own latrines tend to build expensive ones with very large pits or cemented tanks, so the latrines that are socially acceptable are out of reach for many poor households.

WHO and UNICEF's 2014 update estimated that about 597 million people in India still defecated in the open, more than in any other country (World Health Organization [WHO] & United Nations Children's Fund [UNICEF], 2014). Governments have used slogans such as "Toilet first, temple later", "No toilet, no bride" and "Practising good hygiene is as important as performing good puja" to raise awareness, but open defecation remains widespread.

## State Case Studies: Telangana and Bihar

The Swachh Bharat Mission is India's national sanitation programme, aimed at eliminating open defecation by promoting the construction and use of latrines in rural and urban areas. States progressed at different rates. Household survey evidence and research suggest that Telangana emerged as one of the best-performing states, with greater access to latrines and lower open defecation, whereas Bihar continued to face challenges because residents often did not use the sanitation facilities (Palabindela et al., 2020; Jain et al., 2020).

### *Telangana: A Top-Performing State*

A community-based cross-sectional study in rural Telangana found that most respondents had sanitary latrines and only 18% practised open defecation. About 90% of latrines were private toilets that met structural standards, including pucca construction, lighting and ventilation. The main advantage households reported from home toilets was safety. Qualitative and policy research attributes the state's implementation of the Mission and related programmes to its commitment to capacity building and local accountability systems, supported by political leaders and administrative officials. States that set clear objectives and involved local communities were more successful in getting households to install and use latrines (Palabindela et al., 2020).

Social norms against open defecation emerged alongside the increase in latrine use, and the two reinforced each other: where norms favoured latrines, households built and used them, which reduced public defecation. Telangana's stronger economic growth, its rural infrastructure projects and its administration of the Mission are also reported to have created conditions that made it easier for households to use sanitary facilities. Telangana's experience shows that sanitation needs more than toilet construction: social structures and administrative systems must establish rules that support lasting use (Palabindela et al., 2020).

### *Bihar: Persistent Challenges*

Research shows that rural Bihar continues to face deep sanitation problems despite the Mission. An ethnographic study of rural Bihar villages found that the Mission raised household latrine coverage, but that there was a significant gap between toilets built and toilets actually used. Socio-structural and political factors created obstacles to sanitation efforts (Jain et al., 2020).

The study identifies several reasons for Bihar's weak progress. Local citizens saw sanitation programmes as top-down initiatives without adequate local representation. Distrust of government development efforts, and a belief that development favoured certain groups, reduced interest in using the latrines that were available. Government subsidies enabled households to build toilets, but problems in subsidy design led to disbursements that were inadequate or irregular and that recipients perceived as unfair. This lack of trust weakened commitment to sanitation programmes.

### *Comparing Telangana and Bihar*

The comparison offers three lessons about what determines sanitation success. First, success depends on implementation: states that use Mission resources well, with community participation, proper monitoring and behaviour change communication, achieve better sanitation outcomes. Second, community standards, meaning shared expectations about latrine use that people follow in practice, create a stronger system that encourages use, as in Telangana. Third, in Bihar, distrust and weak subsidy mechanisms undermined the promise of the programme, and many households continued to defecate in the open (Palabindela et al., 2020; Jain et al., 2020).

Toilet construction alone cannot end open defecation: habits must change, water must be available and government systems must run without interruption. Telangana's success is associated with better implementation, community adoption and government backing, while

Bihar's high open defecation is associated with social and political problems, low trust in the programme and weak translation of infrastructure into everyday use. Effective sanitation systems therefore need latrines, initiatives to change behaviour and promote fairness, and effective governance together (Palabindela et al., 2020; Jain et al., 2020).

## Empirical Evidence

This section uses NSS data to examine the determinants of access to improved sanitation in Bihar and Telangana, the low-performing and top-performing states respectively. The determinants analysed are (1) access to latrines, (2) reasons for not using latrines and (3) availability of water in latrines, across social group, gender and sector (rural or urban).

The NSS data show that 87.98% of women in Bihar and 86.16% in Telangana use latrines exclusively within their households, so women's access to latrines is high in both the low-performing and the top-performing state.

Reasons for not using latrines differ across social groups. Among Scheduled Tribes (ST) in Bihar, the primary reason is personal preference, and around 45% of the ST community do not use a latrine. This suggests that the Mission has not yet brought about behavioural change in this community, which works against social inclusion in the state. The second most common reason is affordability: around 13% of Scheduled Caste (SC) households in Bihar cannot afford a latrine, compared with only 1% of Other Backward Classes (OBC) households. This points to wide disparities in access across communities in Bihar, which may reflect the caste hierarchy discussed above.

In rural Telangana, the major reasons ST households give for not using a latrine are affordability and other reasons, and 38% of Scheduled Tribes cannot afford a latrine. In urban Telangana the reason is not specified. For SC, OBC and other groups, neither affordability nor a shortage of latrines explains lack of access in rural or urban areas. By contrast, in Bihar around 60% of ST, 5% of SC and 7% of OBC respondents give personal preference as the reason for not using a latrine, while Telangana reports negligible responses of this kind. Personal preference is therefore a far more common reason in Bihar than in Telangana, which is consistent with the role of prevailing norms discussed earlier.

The data also show the highest reported receipt of benefits under the Swachh Bharat Mission among social groups. In Telangana, these are 78% for SC and 77% for OBC in rural areas, and 87% for both SC and OBC in urban areas. In Bihar, they are 84% for SC and 83.6% for OBC in rural areas, and 84% for both in urban areas. Reported receipt of benefits in rural Bihar is therefore higher than in rural Telangana. This does not mean the policy is working poorly in Telangana. It more likely reflects Telangana's sturdier existing sanitation infrastructure and better standard of living, so that fewer households needed support, whereas Bihar still lacks the infrastructure to make its villages and households open defecation free. The data thus point to differences in standards of living between the two states, which affect access to latrines as a basic sanitation requirement.

## Overall Analysis

Availability of water in latrines appears to be an important factor in whether latrines are used, in states where schemes such as the Swachh Bharat Mission, the Atal Mission for Rejuvenation and Urban Transformation (AMRUT), the Smart Cities Mission and other central and state schemes are in place. Results vary with social group, gender and sector, showing how these schemes reach different groups in urban and rural spaces.

Water availability in latrines is 1.7 times higher for Scheduled Castes than for Scheduled Tribes. For OBC it is 2 times higher, and for other groups 3.2 times higher, than for Scheduled Tribes.

By gender, males and females have appropriate availability of water in latrines, but transgender persons do not: availability for a transgender person is 38% lower than for a male, so a major share of the transgender population lacks this basic facility. These estimates for transgender persons may rest on a small number of observations and should be read with caution.

The main constraints persist in rural areas, where informal factors such as social group, social status, gender, cultural practices and personal preference against using a latrine are prevalent. These contribute to open defecation by reducing the likelihood of accessing a latrine. Water availability also differs sharply by sector: urban areas have 2.3 times more water availability in latrine premises than rural areas.

## Conclusion

This paper shows that the persistence or elimination of open defecation in India cannot be explained by toilet construction alone. It is also linked to beliefs, values, caste and untouchability. Government schemes such as the Swachh Bharat Mission, AMRUT and the Smart Cities Mission aimed to expand sanitation infrastructure, but the empirical findings and state-level comparison here show that access to a latrine does not necessarily mean sustained use. Sanitation outcomes are shaped by structural inequalities, governance capacity, behavioural norms and the reliability of complementary services such as water supply.

The contrast between Telangana and Bihar shows how effective governance, administrative coordination and accountability can help reduce open defecation. Telangana has both higher latrine installation and higher use. In Telangana, personal preference is a negligible reason for not using a latrine, although affordability remains a barrier for Scheduled Tribes (38%). In Bihar, personal preference is a common reason for non-use among Scheduled Tribes, and reported receipt of Mission benefits is high even though outcomes are poorer, which suggests that receiving a subsidy does not guarantee effective implementation. This is consistent with the literature reviewed above on mistrust, weak monitoring and top-down governance in Bihar. Telangana's stronger outcomes can be linked to administrative coordination, local accountability and social mobilisation. In Telangana, infrastructure, governance and social change intersect to produce a sanitation transformation; in Bihar, structural inequality and weak governance mean that building toilets is not enough.

## References


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